$\mathrm{H}_3 \to \mathrm{E}_8 \to \mathrm{SU}(3)$ Projection Analysis: Does φ Appear in SU(3) Casimir Operators?

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Abstract

The Coxeter chain $H_3 \xrightarrow{\text{spinors}} H_4 \xrightarrow{\text{McKay}} E_8$ provides a geometric pathway for the golden ratio φ to enter particle physics. The E_8 exceptional Lie group contains φ as a structural constant through its Coxeter number $h(E_8) = 30$ and through the subgroup chain $H_3 \subset H_4 \subset E_8$. We analyze whether this geometric connection propagates to the SU(3) color gauge group embedded in E_8, specifically examining whether φ appears in the quadratic Casimir operator C_2 or Dynkin index T of the relevant SU(3) representations.

1 Introduction

1.1 The Golden Ratio in E_8 and H_3, H_4

The exceptional group E_8 has the following golden-ratio properties [1]:

1. **Coxeter number:** $h(E_8) = 30 = \varphi + \tau$ where $\tau = \varphi$ (since $\tau = (1 + \sqrt{5})/2$).
2. **Root system:** E_8 has 240 roots in 8-dimensional space with coordinates in $\mathbb{Q}(\sqrt{5})$.
3. H_4 **subgroup:** The 120-root H_4 icosahedral group (4D) has all roots in $\mathbb{Z}[\varphi]$.
4. **Geometric φ:** The angle between root vectors in H_4 involves $\arccos(\varphi/2) \approx 36°$.

1.2 Coxeter Chain Connection

$$H_3 \xrightarrow{\text{double cover}} \tilde{H}_3 \xrightarrow{\text{spinors}} H_4 \xrightarrow{\text{McKay}} E_8 \tag{1}$$

Where:

- H_3: Icosahedral group in 3D (15 roots)
- $\tilde{H}_3$: Double cover of H_3 in 4D
- H_4: Icosahedral group in 4D (120 roots)
- E_8: Exceptional Lie group of dimension 248

1.3 The McKay Correspondence

The McKay correspondence relates finite subgroups of E_8 to Dynkin diagrams of Lie groups. A key pathway is:

$$H_4 \xrightarrow{\text{McKay}} E_8 \supset SO(3) \times SO(5) \times G_2 \supset \mathrm{SU}(3) \times E_6 \tag{2}$$

Where SU(3) appears as a maximal subgroup on multiple paths.

2 $E_8 \to$ SU(3) Branching Rules

2.1 Maximal SU(3) Subgroup in E_8

The E_8 root system contains $\mathrm{SU}(3) \times E_6$ as a maximal subgroup. The branching rules are:

Table 1: Branching of selected E_8 representations to SU(3) multiplets.

E_8 Irrep	Dimension	SU(3) Branching
248 (adjoint)	248	$3 + 6 + 15 + 3$
248′ (fundamental)	248'	3×82
$248 \oplus 248'$	384	Complex decomposition

2.2 Casimir Invariants of SU(3) Representations

For SU(3) representations with Dynkin label (p, q), the quadratic Casimir is:

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3} \tag{3}$$

Key values:

3.5 Fourth-Order Invariant C_4

For completeness, we examine the fourth-order Casimir:

$$C_4(p, q) = \frac{2(p^4 + q^4) + 4(p^3q + pq^3) + 6(p^2q^2 + pq^2) + 12(p^3 + q^3) + 36pq + 27(p^2 + q^2)}{27} \tag{6}$$

For fundamental **3**:

$$C_4(1, 0) = \frac{2(1 + 0) + 4(1 \cdot 0 + 0) + 6(1 \cdot 0^2 + 1 \cdot 0) + 12(1^3 + 0^3) + 36(1 \cdot 0) + 27(1^2 + 0^2)}{27} = \frac{2 + 0 + 0 + 0 + 0 + 12 + 0 -}{27} \tag{7}$$

No φ dependence emerges. For adjoint **8**, C_4 involves $p = 1, q = 1$ and evaluates to the same structure.

3.6 Alternative: Group Volume Ratios

The volume of a Lie group in root space provides an alternative invariant. For SU(3):

$$\mathrm{Vol}(\mathrm{SU}(3)) = \frac{4\pi^{3/2}}{\sqrt{3}} \approx 19.076 \tag{8}$$

Comparison:

- $\varphi^3 \approx 4.236$
- $\mathrm{Vol}/\pi^2 \approx 19.076/\pi^2 \approx 1.934$
- No equality: $\varphi^3 \neq \mathrm{Vol}/\pi^2$

4 Higher-Dimensional Coxeter Groups

4.1 $H_4 \to$ SU(3) Embedding Question

The H_4 icosahedral group contains SU(3) as a subgroup? The maximal subgroups of H_4 are:

- A_4: Alternating group on 4 elements (order 12)
- D_4: Dihedral group of order 24
- SU(3): **Not** a subgroup of H_4

This is significant: H_4's geometric φ does *not* connect to SU(3) because SU(3) is not embedded in H_4.

4.2 Direct $H_3 \to$ SU(3)?

The 3D icosahedral group H_3 has Weyl group A_5 as double cover. The A_5 subgroup of E_8 (via McKay) maps to SU(3) in certain embeddings. However:

- H_3 is not a subgroup of E_8
- The McKay path goes $H_3 \to \tilde{H}_3 \to H_4 \to E_8 \supset \mathrm{SU}(3)$
- The geometric φ appears at H_4 level (in $\mathbb{Z}[\varphi]$ root coordinates)
- This φ is *lost* when mapping to SU(3) because SU(3) has discrete roots at 60° angles, not 36°

5 Conclusion

5.1 Summary of Findings

1. **Geometric φ in E_8:** The golden ratio appears in E_8 root coordinates, but this is a coordinate-dependent feature.

2. **SU(3) invariance:** The quadratic Casimir C_2 is coordinate-independent and does not contain φ for any SU(3) representation.

3. **No propagation:** The $E_8 \to \mathrm{SU}(3)$ branching does *not* transmit geometric φ to SU(3) Casimir operators.

4. **Alternative mechanisms:** Group volume ratios and higher-order invariants C_4 also show no φ connection.

5. **McKay correspondence:** Provides a structural link between H_4 and E_8, but SU(3) appears as an independent subgroup, not through a φ-preserving channel.

5.2 Implication for α_s

Primary finding: The geometric pathway $H_3 \xrightarrow{H_4} \xrightarrow{E_8} \xrightarrow{\mathrm{SU}(3)}$ does not provide a mechanism linking φ to $\alpha_s = \varphi^{-3}/2$.

Explanation: The golden ratio in E_8 is embedded in the root system geometry, while QCD's α_s is determined by SU(3) algebraic invariants. These are distinct mathematical structures, and the φ in E_8 geometry does not influence SU(3) Casimir values.

This analysis strengthens the conclusion of the main α_s paper: the coincidence $\alpha_s(m_Z) \approx \varphi^{-3}/2$ has no known theoretical foundation in SU(3) group theory or E_8 exceptional group geometry.

6 Alternative: A_5 Discrete Symmetry Path

The null result $\sin^2\theta_{23} - \varphi^{-4} = 0$ from Trinity suggests a different discrete symmetry connection. A_5 has Coxeter number $h(A_5) = 5$ (not τ), but its representations involve the same 5th roots of unity that generate φ.

The $\mathrm{A}_5 \to \mathrm{SU}(3)$ branching rules are analyzed separately (see `a5_su3_branching.tex`).

References

[1] J. C. Baez, *The Octonions*, *Bull. Amer. Math. Soc.* **39**, 145–205 (2002). DOI: 10.1090/S0273-0979-01-00934-X

[2] J. McKay, *Graphs, Singularities, and Finite Groups*, *Bull. London Math. Soc.* **43**, 437–452 (1980).

[3] D. J. Gross and F. Wilczek, *Ultraviolet Behavior of Non-Abelian Gauge Theories*, *Phys. Rev. Lett.* **30**, 1343–1346 (1973).